Mustapha M. Strachan

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EDUCATION

University of Central Florida Expected Spring 2028
Bachelor of Science in Computer Science | GPA: 3.75 *Orlando, FL*

- Relevant Coursework: Data Structures & Algorithms, Object-Oriented Programming, Computer Organization, Systems Software, Discrete Structures, Calculus I & II
- Honors & Involvement: Dean's List, Honor Roll, Knight Hacks, National Society of Black Engineers, UCF DARTS

EXPERIENCE

Quickstart SimJam Intern June 2026 - July 2026
UCF Institute for Simulation and Training (METIL) *Orlando, FL*

- Collaborated on a QIF-to-Unity pipeline that converts QIF XML geometry and topology into selectable Unity mesh data
- Implemented and debugged C++ parsing and OpenCascade translation components for QIF curves, surfaces, and B-rep geometry
- Used Git, GitHub, Visual Studio, and Unity to integrate native C++ processing with a C# importer workflow
- Collaborated with teammates across geometry import, feature selection, and product manufacturing information display

SOAR Program Peer Mentor May 2026 - Present
University of Central Florida *Orlando, FL*

- Mentor incoming students and support their transition to university academics, resources, and campus life
- Facilitate student engagement activities, one-on-one mentoring sessions, and professional communication with students and staff
- Share program updates through Microsoft Teams and email while tracking student participation and attendance

PROJECTS

Sport IQ | Python, FastAPI, Next.js, TypeScript, Machine Learning, Supabase

- Built a full-stack sports prediction platform for MLB, NBA and NFL games using historical team and game data to generate pregame win probabilities and confidence scores
- Developed FastAPI services and a Next.js/TypeScript interface for daily matchups, model explanations, leaderboards, user profiles, favorites, and prediction history
- Evaluated predictions with accuracy and Brier score while maintaining a results history for performance tracking
- Deployed the application on Vercel with Supabase authentication/database and automated daily prediction workflows; transitioned the product into a free social sports platform

AI DJ | Python, FastAPI, Librosa, NumPy, Audio Processing

- Built a FastAPI AI DJ system that analyzes WAV/MP3 tracks and scores compatibility by tempo, key, energy, and structure
- Generates mix plans and playable WAV audio with beat-aware crossfades, phrase-based cue points, normalization, and transition SFX
- Added a browser live deck with queue controls, track search, and learned DJ style profiles from reference mixes

AI Civilization | Python, FastAPI, Next.js, TypeScript, Multi-Agent Systems

- Built a persistent civilization simulator where autonomous citizens work, form relationships and families, age, and leave histories
- Developed agent brains with personality traits, needs, emotions, goals, memories, utility-scored actions, and explainable decisions
- Created a FastAPI backend and Next.js dashboard with citizen profiles, event feeds, simulation controls, and persistent world state

LEADERSHIP & ACTIVITIES

Campus Involvement: Knight Hacks, National Society of Black Engineers, Alpha Phi Alpha Fraternity Incorporated, UCF DARTS, NAACP, UCF Intramural Sports
Honors: Dean's List and UCF Honor Roll
Awards: 2nd Place overall Knight Hacks hackathon, Best Use of Hardware SharkByte hackathon, Strong AI Scholarship

TECHNICAL SKILLS

Languages: C, C++, C#, Python, Java, TypeScript, HTML/CSS
Frameworks: FastAPI, Next.js, React
Developer Tools: Git, GitHub, Docker, Unity, OpenCascade, Visual Studio, VS Code, Linux
Concepts: Data Structures & Algorithms, Object-Oriented Programming, REST APIs, Machine Learning, XML Parsing, 3D Mesh Processing